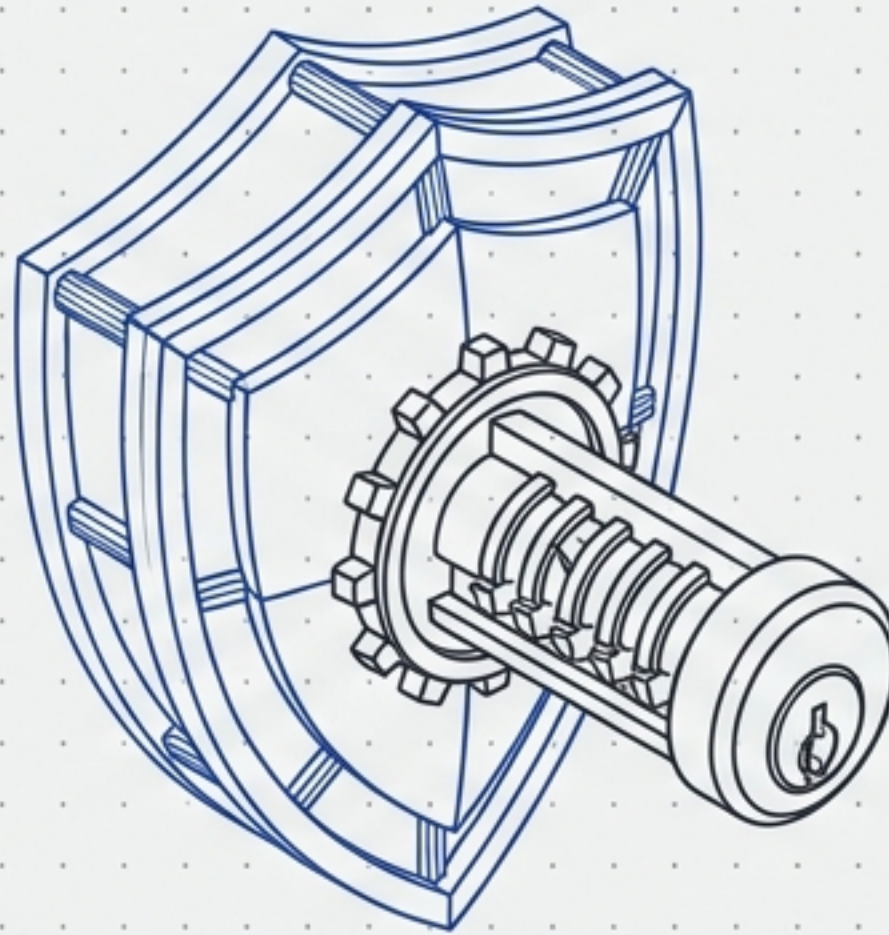


Assets, Threats, and Vulnerabilities

A definitive blueprint of the Google Cybersecurity Professional Certificate curriculum.

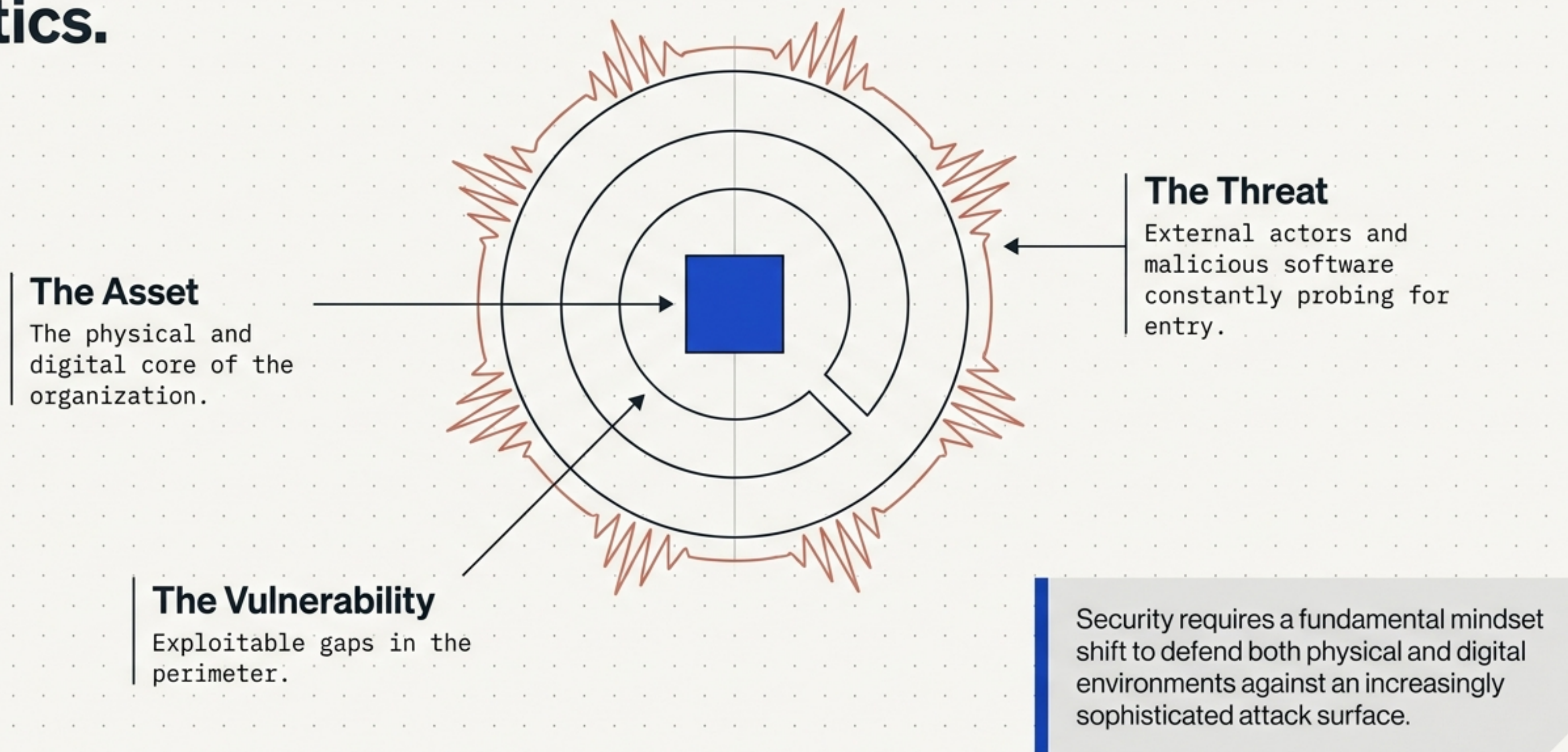


Beginner Level: No prior
experience required

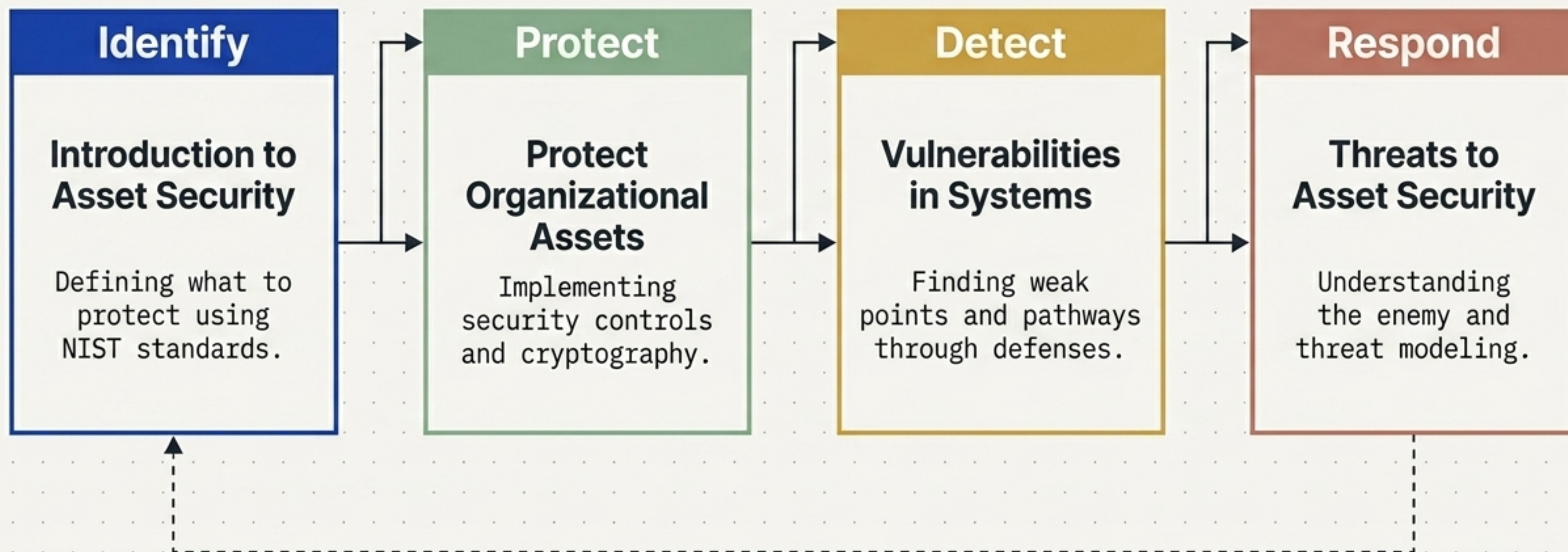
Flexible Schedule:
10 hours/week

Validation:
4.8 Star Rating

The modern imperative is protecting organizational assets from ever-evolving tactics.

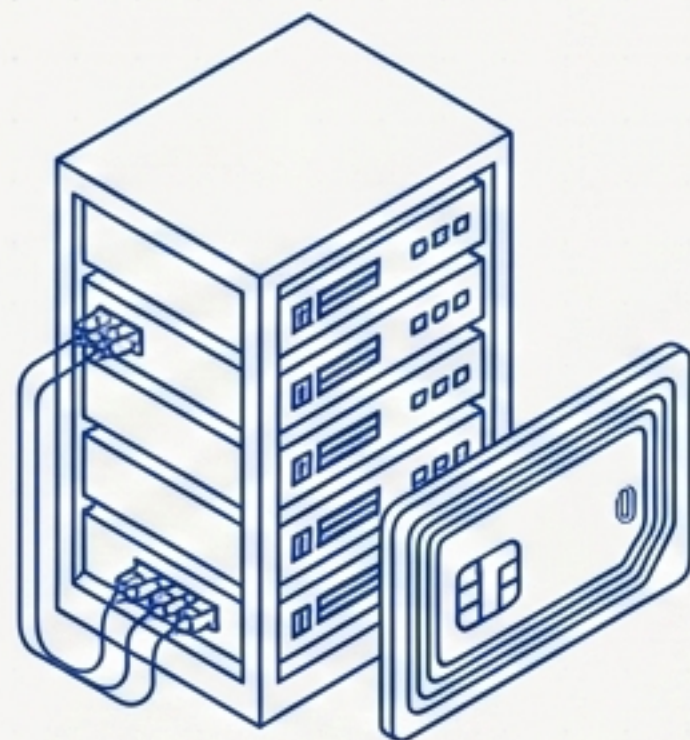


The Defender's Journey: Building a complete security architecture.



Security starts with rigorous asset classification.

Physical World

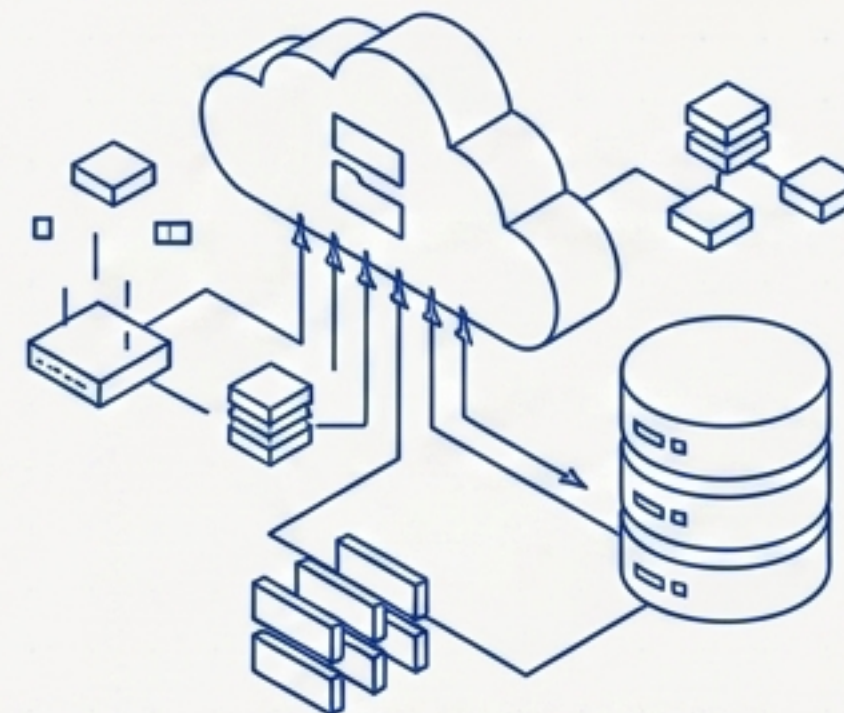


Hardware, facilities, and physical access points.

The NIST Cybersecurity Framework (CSF)

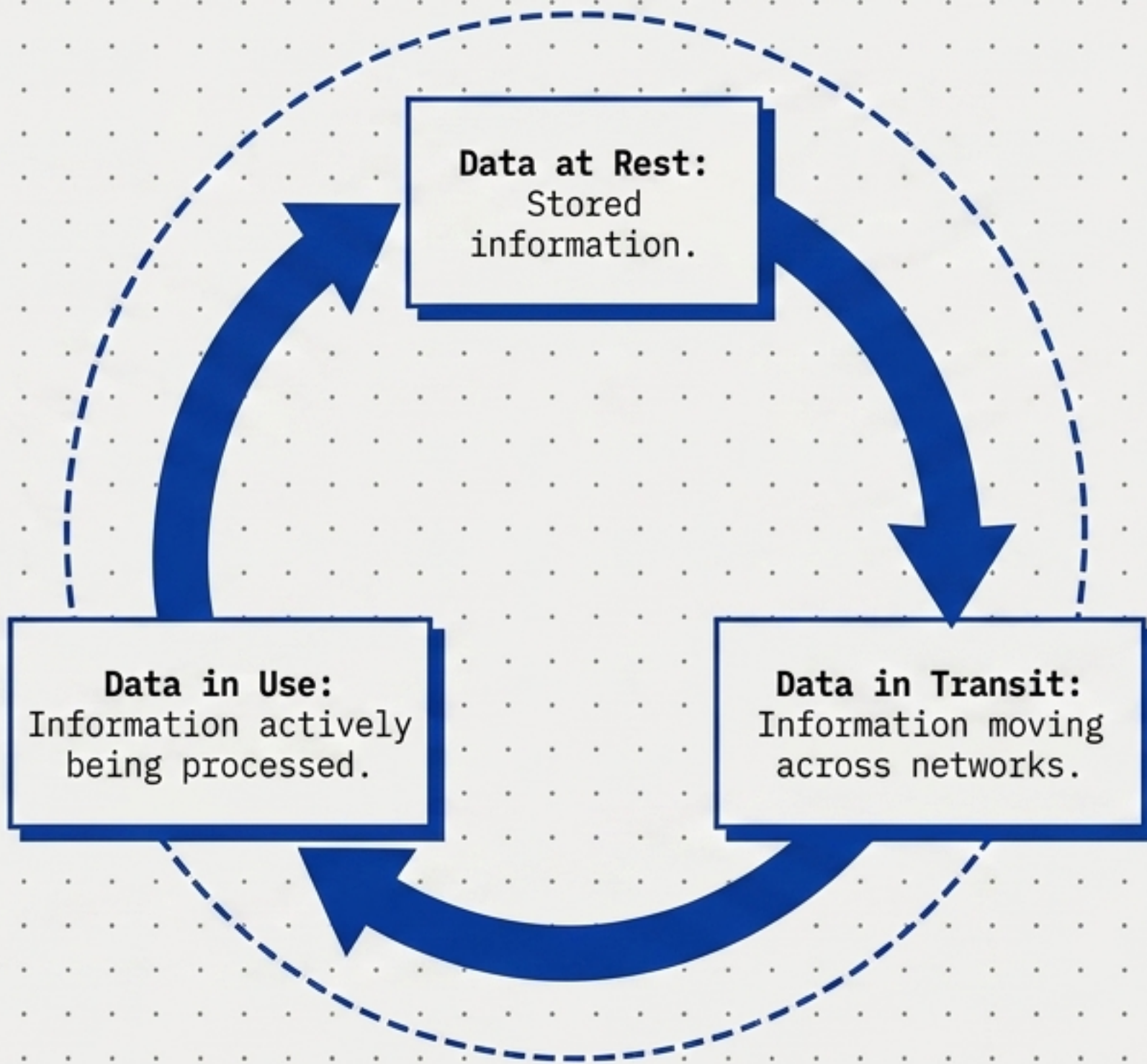
The industry gold standard used to score risks based on likelihood and severity, guiding organizations in managing complex cybersecurity risks.

Digital World



Cloud environments, PII (Personally Identifiable Information), and organizational data.

Applying the mechanics of cryptography to the data lifecycle



Security Mechanisms

Encryption

Symmetric (single key)
vs.
Asymmetric
(public/private key
infrastructure).

Hashing

Ensuring data integrity
and non-repudiation.

Securing entry points through Identity and Access Management

The Access Framework



Authentication

What it is: Verifying who the user claims to be.

Key Mechanisms: Single Sign-On (SSO), Multi-Factor Authentication (MFA).



Authorization

What it is: Determining what the user is allowed to do.

Core Principle: Principle of Least Privilege.

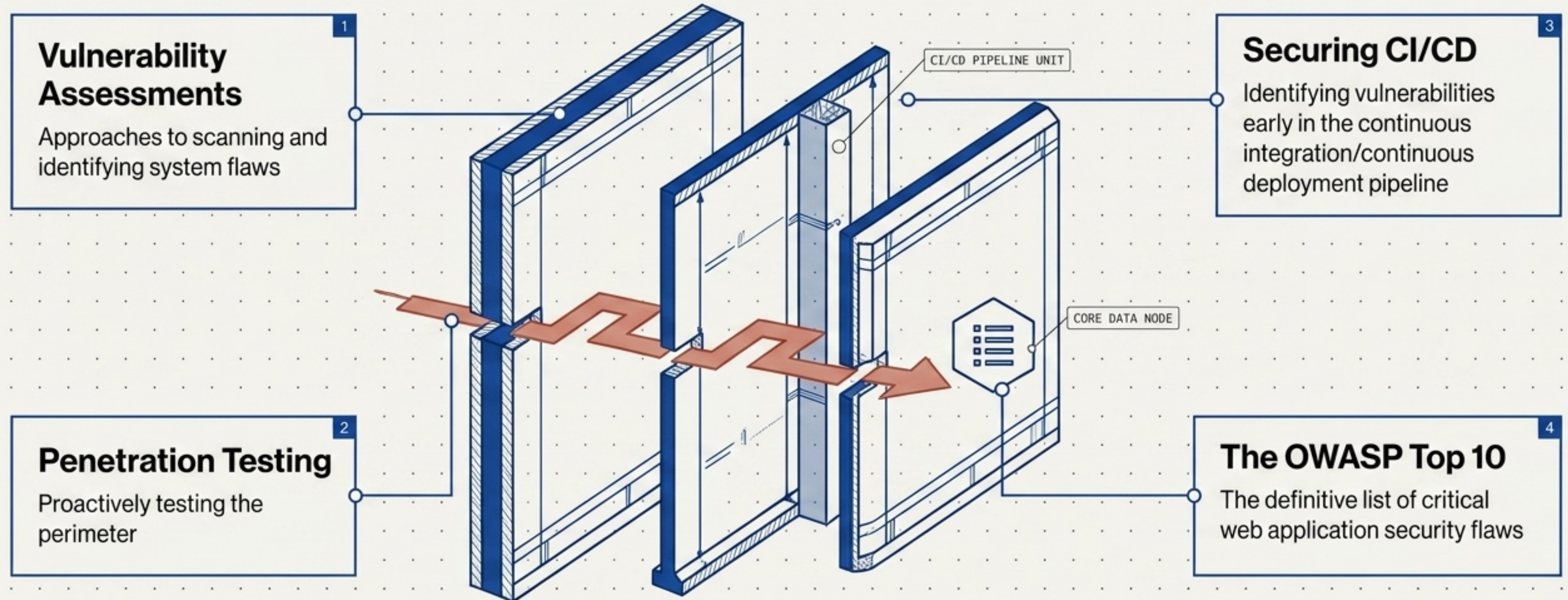


Auditing (Accounting)

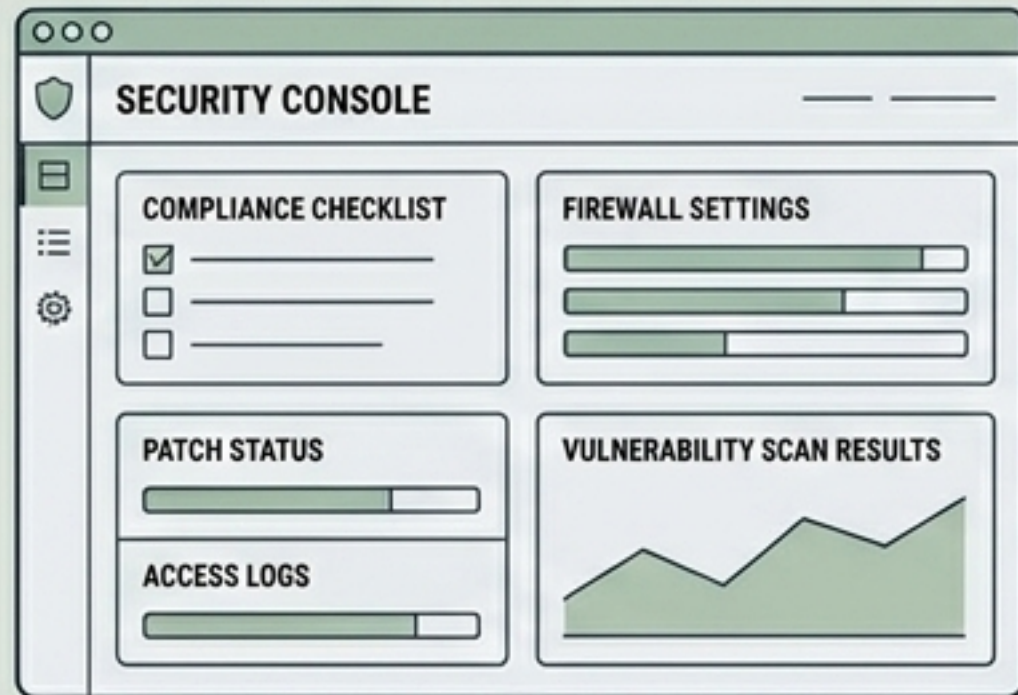
What it is: Tracking user activity and access history.

Why it matters: Ensuring accountability and detecting anomalous behavior.

Uncovering system vulnerabilities through defense in depth



You cannot protect an asset without adopting an attacker mindset.



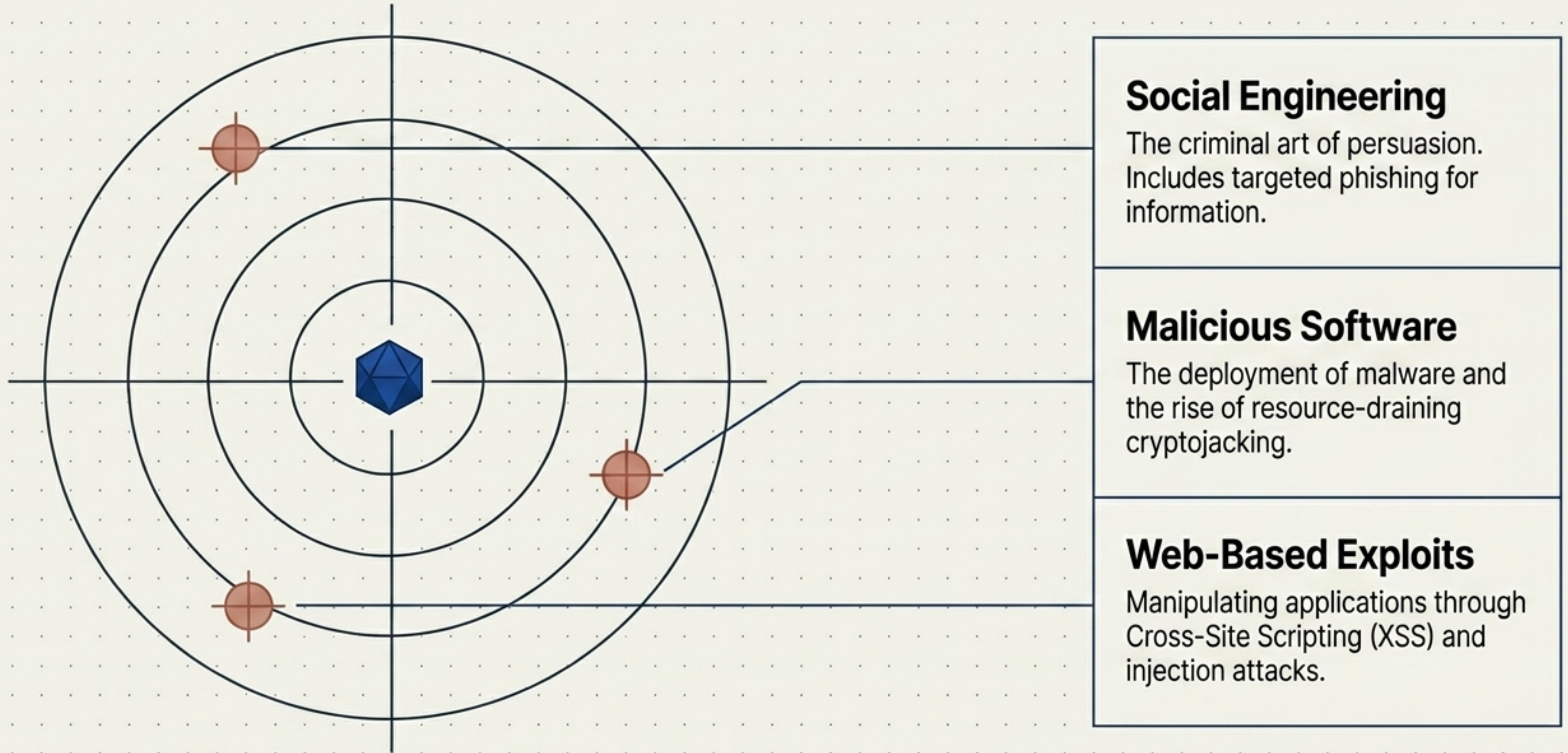
Focuses on fortifying entry points, applying patches, and managing vulnerability scanners.



Focuses on Open Source Intelligence (OSINT), identifying attack vectors (e.g., a dropped USB drive), and exploiting brute-force opportunities.

Effective security requires seeing the exploitable gaps in your own database before the threat actor does.

Mapping the modern threat landscape.



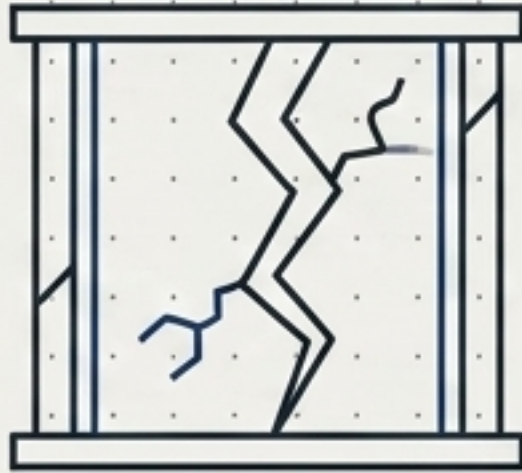
Proactive security via the PASTA framework.

The Process for Attack Simulation and Threat Analysis.



Security professionals use PASTA to stay one step ahead of security breaches.

Distinguishing the flaw from the danger.



Vulnerabilities

- **Definition:** Inherent weaknesses or flaws in a system.
- **Examples:** Unpatched software, exposed databases, OWASP Top 10 flaws.
- **The Analogy:** A cracked window.



Threats

- **Definition:** External actors or elements that exploit those flaws.
- **Examples:** Cybercriminals, Malware, Social Engineering.
- **The Analogy:** The burglar.

The Cybersecurity Toolbelt:

Fluency in industry frameworks.

NIST

National Institute of Standards and Technology. Used for overall risk and asset management guidelines.

MITRE ATT&CK

Used for identifying and understanding common vulnerability exposures and adversary tactics.

OWASP

Open Worldwide Application Security Project. The benchmark for identifying top web application flaws.

PASTA

The step-by-step methodology for attack simulation and threat modeling.

Acquiring highly actionable, job-ready capabilities.



Cyber Threat
Intelligence



Cryptography
& Hashing



Identity & Access
Management (IAM)



Risk Management
Frameworks



Security Controls
Deployment



Vulnerability
Assessments



Malware Protection
Strategies



Data Lifecycle
Management

Validated by a global community of defenders.

352,208

Already Enrolled

4.8

Star Rating (4,114 reviews)

98%

Learner Satisfaction

“To be able to take courses at my own pace and rhythm has been an amazing experience.”

– Felipe M.

“I directly applied the concepts and skills I learned from my courses to an exciting new project at work.”

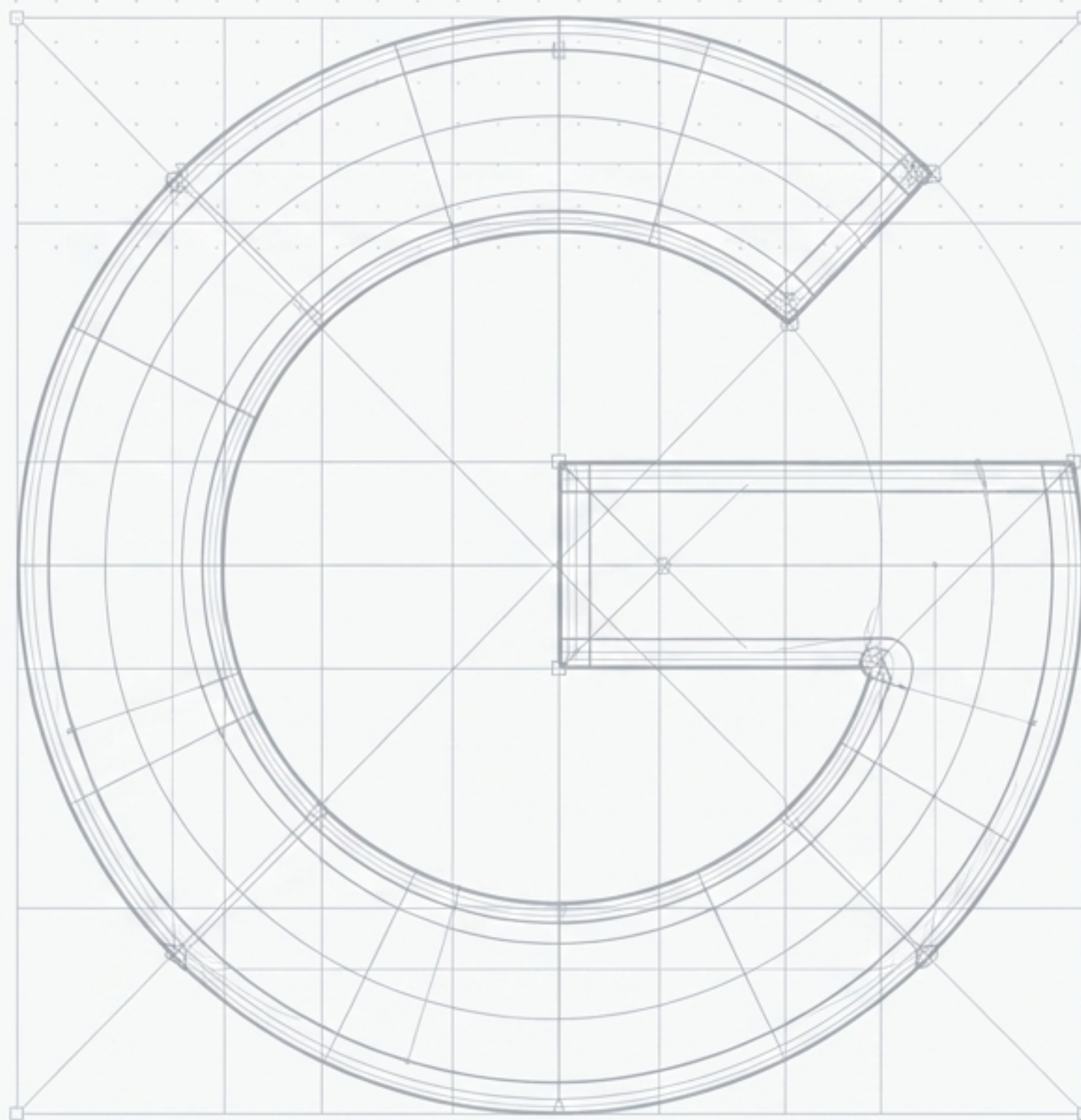
– Jennifer J.



Secure the perimeter. Advance your career.

Begin Assets, Threats, and Vulnerabilities—Course 5 of the Google Cybersecurity Professional Certificate

Enroll Now



Shareable certificate upon completion | Available for Individuals via Coursera Plus | Deployable for teams via Coursera for Business